VCD / 111022 SYBSC SEM III BOTANY II 100 MARKS 3 hours Ex.

Please check whether you have got the right question paper

N.B.

1. **All questions are compulsory.**
2. **Figures to right indicate full marks.**
3. **Draw neat and labeled diagrams wherever necessary.**

Q.1 A Choose the correct option from the following **10**

1. Where is DNA present in the eukaryotic cells?
   a. Inside the nucleus c. With other cellular contents
   b. Inside the ribosomes d. Not present
2. The process of DNA replication is affected by an enzyme known as_________.
   a. Mutase b. Ligase c. Polymerase d. Ribonuclease
3. Genetic information stored in mRNA is translated to polypeptide by ___________
   a. Ribosome b. Nucleus c. Endoplasmic reticulum d. Golgi apparatus
4. ______ is the type of aberrations that involves the loss of a chromosomal segment.
   a. duplication b. deletion c. translocation d. inversion
5. Inheritance of streptomycin resistance or sensitivity in Chlamydomonas is an example of _______ inheritance.
   a. uniparental b. lysosome c. biparental d. nuclear
6. __________ is an example of monoecious plant
   a. maize b. rose c. mango d. papaya
7. __________ is an example of heterogametic females which shows the ZZ-ZW mechanism of sex determination.
   a. birds b. grasshopper c. butterflies d. humans
8. Super coiling of single-stranded DNA is prevented by _______________.
   a. DNA Polymerase b. Ligase c. SSBs d. Gyrase
9. During replication, Okazaki fragments elongate on the_____ ________ strand.
   a. Leading b. Lagging c. Dispersive d. None of the above
10. The fragments of DNA are joined together by ________ enzymes.
    a. Endonuclease b. DNA polymerase c. Primase d. Ligase

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Q.1 B Answer the following in one /two sentences each **10**

a. What is the function of DNA?
b. How are the two helical strands wounded?
c. What is translocation? Give its three types.
d. What is Barr body?
e. Give the role of DNA polymerase.

Q.2 Answer *any two* from the following **20**

1. Explain the function and biogenesis of Mitochondria.
2. What are peroxisomes? Explain their structure, function, and biogenesis.
3. What are Glyoxysomes? Explain their structure, function, and biogenesis.
4. Give details on mitosis.

Q.3 Answer *any two* from the following **20**

1. Explain the mechanism behind the following crosses
   a. XX-XY
   b. XX-XO.
2. Concerning chromosomal aberrations give the types of duplication and inversion.
3. Explain colorblindness Genetic Inheritance if
   a. the father has colorblindness and the mother is unaffected.
   b. the mother carries the altered gene causing colorblindness (carrier) and the father is unaffected.
4. Carry out the following crosses to obtain percentages of progenies.
   a. Carrier female and Hemophilia male.
   b. Carrier female and normal male.

Q.4 Answer *any two* from the following **20**

1. Explain the DNA Replication in Eukaryotes with a neat labeled diagram.
2. How did Messelson and Stahl demonstrate DNA Replication? Add a note on its importance.
3. Give the molecular mechanism of protein synthesis in prokaryotes with a neat labeled diagram.
4. Describe the two major steps in mRNA processing?

Q.5 Write short notes on *any four* from the following **20**

a. t-RNA.
b. Significance of Meiosis.
c. Genic balance theory in *Drosophila*.
d. Sex- influenced traits.
e. Models of replication.
f. Enzymes in DNA replication.

XXXXXX All the best XXXXX